DHA Suffa University

Final Year Project

Emotion Aware Emergency Dispatch System (EAEDS)

S31F25

FYP Proposal

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Document Sign off Sheet

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04/1/2026	1.0	Initial document	Abdullah Shaikh,Syed Murtaza Hassan,Syed Muhammad Baqar,Tahir Mehdi

Definition of Terms, Acronyms, and Abbreviations

Terms	Description
EAEDS	Emotion Aware Emergency Dispatch System, which is a system designed to analyze the emotions of callers and help dispatchers in their work.
LLM	A model that is train on alot of data like whole world data like chat gpt,gemini and many more.
Whisper	an open-source ASR model created by OpenAI that is used for converting speech into text.
Latency	the time it takes for a system to respond to an action.
GPU	Graphics Processing Unit, specialized hardware that speeds up AI calculations.
FastAPI	Python framework used in our project backend for creating apis.
Next.js	React framework used for frontend in our project.
SQLite	Database for storing calls and other works
Dashboard	the main page where live call information, emotions, and location of caller is displayed. In our project it is called midnight dashboard
JSON	JavaScript Object Notation. A structured data format used to store transcripts and emotion data.
UI/UX	Design of how our dashboard looks
SQLite	Database for storing calls and other works
Call Session	A single emergency interaction that includes audio, transcript, emotion data, and metadata such as time and location.
BERT	Bidirectional Encoder Representations a model used for understanding natural language and analyzing emotions.
Whisper	an open-source ASR model created by OpenAI that is used for converting speech into text.
Open Smile	an open-source tool that extracts acoustic and emotional features from audio signals.

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1 Introduction

1.1 Problem Statement

Global emergency services are facing critical staffing shortages, with reports showing an 82% deficit in 911 call centers. Human operators are burdened by high call volumes and high-stress environments, leading to:
Increased response times due to cognitive overload.
Human errors in information recording and dispatch decisions.
Lack of context, as traditional systems rely on audio-only communication without automated sentiment analysis.

1.2 Product Position Statement:

The Emotion-Aware Emergency Dispatch System (EAEDS) is an AI-powered dispatch operator which is constructed for essential emergency response centers. Not like a traditional CAD (Computer Aided Dispatch) systems, EAEDS reviews the audio and emotion of the caller in real-time with the use of local GPU processing, which provides controllers with immediate transcription, emotional telemetry, and triage recommendations without depending on external cloud services.

1.3 Project Motivation and Background

Emergency dispatch systems are considered as the first line of defense in safety of public. However, current setups are no longer used, depending a much on manual entry and human reasoning under too much pressure. They are encouraged by nowadays improvements and updates in local AI inference (Quantized LLMs), the main aim of this project is to fill the gap with the help of creating a smart operator that usually listens, understands, and thinks at the same time as the human operator and convert raw audio into actionable logistical data immediately within a seconds.

1.4 Objectives

Reduce Triage Time: Automate the collection of details (Name, Location, Type of Emergency) to speed up dispatch.
Enhance Situational Awareness: Provide operators with live visual indicators of the caller's emotional state (Panic vs. Calm).
Ensure Data Privacy & Reliability: Operate entirely on local hardware (NVIDIA GPU), eliminating privacy concerns and internet dependency regarding cloud APIs.
Operational Resilience: Ensure the interface remains responsive even under heavy load.

1.5 Literature Review and GAP Analysis

Reddy, S., & Kompalli, S. (2022). Emotion Detection using Multimodal Analysis. - Discusses using audio and text for emotion detection but relies on cloud APIs which introduces latency.
Huang, J., et al. (2020). Whisper and ASR Systems. - Establishes the accuracy of Whisper but highlights the computational cost for real-time usage.

GAP Analysis:

Most nowadays existing "Smart 911" solutions depends on cloud-based APIs (Google Cloud Speech, AWS Transcribe). This gives introducion to two critical gaps:
Latency: Internet round-trip times are not acceptable for split-second emergency decisions.
Privacy/Security: Sending emergency call audio to third-party clouds goes against the harsh data sovereignty laws.
Our Solution (EAEDS) closes these gaps by processing the entire AI pipeline (Whisper + BERT + LLM) locally on the workstation's GPU, by making sure 0-hop privacy and minimum latency.

2 Project Vision

2.1 Business Case and SWOT Analysis

Business Case:

By deploying EAEDS, emergency centers can manage 30% more calls with the same and stable staff levels by removing data entry confirmation to AI. The clearance of repeated and frequent cloud subscription outlay (API usage fees) results in a important reduction in Operational Expenditure (OpEx) after the initial and beginning hardware commitment.

SWOT Analysis:

Strengths	Weaknesses	Opportunities	Threats
Privacy First: No data leaves the facility (Local GPU).	Hardware cost: Requirement of high-end NVIDIA GPUs (RTX 3060+) is needed.	Government Contracts: High requirement for independent AI solutions.	Model Hallucination: AI may hear wrong and misunderstand addresses which is basically mitigated by human-in-loop.
Low Latency: Rapid and fast decisions without internet delay and decisions are not slowed by internet.	Setup Complexity: It is very hard to install than a web-SaaS.	Expansion: It is a modular design that allows Medical and Fire specifics.	Hardware Failure: It is a one and single point of failure if GPU burns out and stop working.

2.2 Stakeholder Summary

Type	Description	Responsibilities
Primary Users	911 Dispatchers	Monitor dashboard, verify and check AI suggestions, dispatch units.
Secondary Users	First Responders (Police/EMS)	Receive processed and incident reports and location data.
Administrators	System Admins	Controls and maintain GPU servers, improve Docker containers, handle logs.

2.3 User Summary

Name	escription	Responsibilities	keholder
Dispatcher	Trained emergency operator	Manage calls, review AI transcripts, override AI decisions if wrong.	Internal
Administrator	Technical Officer	Ensure the system uptime, handle backups through Docker.	Internal

2.4 Business Objectives and Success Criteria:

Sub-2-Second Latency: Transcription should must appear within 2 seconds of speech.
99.9% Local Uptime: System must operate and work without internet access.
Accuracy: Emotion detection must match with the human consensus >80% of the time.

2.5 Project Risks and Risk Mitigation Plan

Project Risks:

GPU Resource Exhaustion: Processing Whisper, LLM, and BERT simultaneously might crash VRAM.
Model Hallucination: The LLM will might suggest "Fire Dept" for a "Firearm" incident.
Audio Quality: Poor signal Quality will lead to bad transcription.

Risk Mitigation Plan:

Quantization:
Usage of 4-bit quantized models (GGUF/AWQ) to fit 3 distinct models into 8GB VRAM.
Human-in-the-Loop:
The actions are only recommended by the AI dispatcher, the human Dispatcher must click "Dispatch" to confirm it
Fallback Mode:
The system reverts to a standard "Dumb" phone interface immediately if the AI fails.

2.6 Assumptions and Dependencies

Assumptions

The deployment site has secure power and a workstation with an NVIDIA GPU (6GB+ VRAM).
Callers must speak English.

Dependencies:

NVIDIA CUDA Toolkit:It is essential for local acceleration.
Docker:It is essential for containerized deployment.
Mapbox/Leaflet:It is essential for rendering the offline-capable map tiles.

3 Project Scope

The design and development of a fully functional local-first dispatch assistant prototype is covered by the project.

3.1 In Scope

Real-time Speech-to-Text: It is local Whisper inference.
Emotion Detection: It is Acoustic (OpenSMILE) + Semantic (BERT) analysis.
Smart Dispatch Logic: It is Local LLM (Mistral/Phi-3) deciding priority.
Midnight Dashboard: It is Next.js based dark-mode UI.
Local Architecture: It is Full offline capability (except Map tiles initially).

3.2 Out of Scope

PSTN Hardware integration: We use SIP/WebSockets, and do not use physical telephone wires.
Legal Liability: The System is decision-support only.
Multi-language Support: There must be English only for this version.

4 Proposed Methodology

4.1 SDLC Approach (Waterfall/Agile/Spiral)

We are following an Agile Iterative methodology.
1. Phase 1 (MVP): Basic transcription and Dashboard that is completed.
2. Phase 2 (Intelligence): Integrating Local LLM and BERT that is current.
3. Phase 3 (Refinement): Optimization present for latency and UI Polish.

This allows us to check the heavy GPU load early and help in adjusting the model sizes accordingly.

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4.2 Team Role & responsibilities

Ms. Shaila - Project Supervisor
Ms. Mashal Tariq - Project Co-supervisor

Abdullah Shaikh - Team Leader
Responsibilities - Project Management, Dashboard UI (Next.js), SRS/Documentation.

Syed Murtaza Hassan - Team Member
Responsibilities - Backend Architecture (FastAPI), AI Model Integration (Whisper/LLM), GPU Optimization.

Tahir Mehdi - Team Member
Responsibilities - WebSocket Engine, Database Management (SQLite), Real-time Synchronization.

Syed Muhammad Baqar - Team Member
Responsibilities - Frontend Components, Testing, QA & Bug Fixing.

4.3 Requirement Development Methodology

Data Collection:

500+ mock emergency call transcripts are collected together and audio samplesa are also collected to create a fine-tuning dataset.

Analysis and Design:

The system is designed using Object-Oriented principals, and prioritize the Strategy Pattern which helps to swap AI models easily.

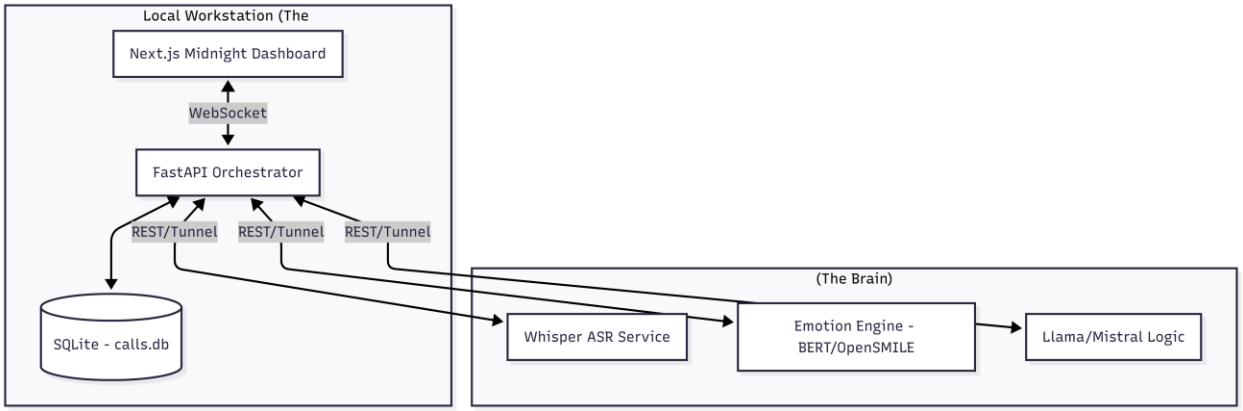
Development and Implementation:

With the use of a microservices approach: Client (Next.js) <-> Server (FastAPI) <-> AI Engine (Python/C++).

Testing:

Unit testing is done for API endpoints and Stress Testing is used for the GPU with concurrent audio streams.

4.4 High level Architecture / Design



5 Project Planning

5.1 Gantt Chart

1	FYP-I	TEAM	Tue 10/07/25	115	Thu 1/29/26	100%	83
0.1	Idea selection	AS,MH,TM,MB	Tue 10/07/25	3	Thu 10/09/25	100%	3
0.1	Super visor selection	AS,MH,TM,MB	Fri 10/10/25	3	Sun 10/12/25	100%	1
0.1	Idea discussion from supervisor	AS,MH,TM,MB	Mon 10/13/25	4	Thu 10/16/25	100%	4
0.1	Idea approval from supervisor	AS,MH,TM,MB	Fri 10/17/25	3	Sun 10/19/25	100%	1
0.1	Final Examination (No Project Work)		Mon 10/20/25	10	Wed 10/29/25	100%	8
0.1	Research for neccesary models,llm and backend framework	AS,MH	Thu 10/30/25	5	Mon 11/03/25	100%	3
0.1	Dataset Finding	AS,MH	Tue 11/04/25	6	Sun 11/09/25	100%	4
0.1	Review from supervisor	AS,MH,TM,MB	Mon 11/10/25	6	Sat 11/15/25	100%	5
0.1	1st Defence of FYP 1		Sun 11/16/25	2	Mon 11/17/25	100%	1
0.1	Data preprocessing (911 calls)	AS,MH	Tue 11/18/25	4	Fri 11/21/25	100%	4
0.1	Discussion with team members about the aproach	AS,MH,TM,MB	Sat 11/22/25	4	Tue 11/25/25	100%	2
0.1	Project Setup(Nextjs/Fast APi)	AS,MH	Wed 11/26/25	7	Tue 12/02/25	100%	5
0.1	Midnight Dashboard UI	AS,MB,MH	Wed 12/03/25	5	Sun 12/07/25	100%	3
0.1	Dataset Finding	AS,MH	Tue 11/04/25	6	Sun 11/09/25	100%	4
0.1	Review from supervisor	AS,MH,TM,MB	Mon 11/10/25	6	Sat 11/15/25	100%	5
0.1	1st Defence of FYP 1		Sun 11/16/25	2	Mon 11/17/25	100%	1
0.1	Data preprocessing (911 calls)	AS,MH	Tue 11/18/25	4	Fri 11/21/25	100%	4
0.1	Discussion with team members about the aproach	AS,MH,TM,MB	Sat 11/22/25	4	Tue 11/25/25	100%	2
0.1	Project Setup(Nextjs/Fast APi)	AS,MH	Wed 11/26/25	7	Tue 12/02/25	100%	5
0.1	Midnight Dashboard UI	AS,MB,MH	Wed 12/03/25	5	Sun 12/07/25	100%	3
0.1	Discussion with supervisor	AS,MH,TM,MB	Mon 12/08/25	3	Wed 12/10/25	100%	3
0.1	Literature Review	AS,MH,TM,MB	Thu 12/11/25	3	Sat 12/13/25	100%	2
0.1	Discuss Literature Review with supervisor	AS,MH,TM,MB	Sun 12/14/25	6	Fri 12/19/25	100%	5
0.1	WebSocket Engine Basic	TM,MH	Sat 12/20/25	3	Mon 12/22/25	100%	1
0.1	Mid Evaluation on first module(Frontend and backend along with cloud testing)	AS,MH,TM,MB	Tue 12/23/25	3	Thu 12/25/25	100%	3
0.1	FastAPI Orchestrator (No AI)	MH	Fri 12/26/25	3	Sun 12/28/25	100%	1
0.1	SQLite Database Connection	TM,AS	Mon 12/29/25	3	Wed 12/31/25	100%	3
0.1	Offline Geocoding (Maps)	AS	Thu 1/01/26	5	Mon 1/05/26	100%	3
0.1	Dashboard Final Polish	AS,MH	Tue 1/06/26	6	Sun 1/11/26	100%	4
0.1	SRS	AS	Mon 1/12/26	3	Wed 1/14/26	100%	3
0.1	SDS	MH	Thu 1/15/26	3	Sat 1/17/26	100%	2
0.1	Project Proposal,Srs,Sds (Changes)	MH	Sun 1/18/26	3	Tue 1/20/26	100%	2

6 Project Requirements

6.1 Software tools requirements:

Languages: TypeScript (Frontend), Python 3.10+ (Backend/AI).
Frameworks: Next.js 14, FastAPI, PyTorch.
AI Libraries: Transformers (Hugging Face), Faster-Whisper, vLLM or Llama.cpp.
Containerization: Docker Desktop.

6.2 Hardware requirements

GPU: NVIDIA RTX 3060 (12GB VRAM) or better.
RAM: 32GB DDR4/DDR5.
Storage: 512GB NVMe SSD (High speed I/O needed for model loading).
OS: Windows 10/11 (with WSL2) or Ubuntu Linux.
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7 Budget/Costing

7.1 Estimated Budgeted Cost of the Project

Total Hours of Development is 400 hrs
Estimated Hardware Cost (GPU Workstation - One Time) is 250,000 Rs
Peripheral Equipment (Headsets/Mics) is 15,000 Rs
Software Licenses (Mapbox/Bootstrap - mostly Open Source) is 0 Rs
Total Estimated Cost is 265,000 Rs (Hardware Asset)

8 Project Deliverables (you may change according to your project methodology)

8.1 Phase I - Alpha Prototype

Basic Dashboard Interface (Next.js).
WebSocket connection is maintained and established.
Dummy data flow verification is done.

8.2 Phase II - Beta Prototype

Integration of Local Whisper (STT) and OpenSMILE.
Live audio transcription pipeline is functioning.
Basic Emotion Classification (Text-based).

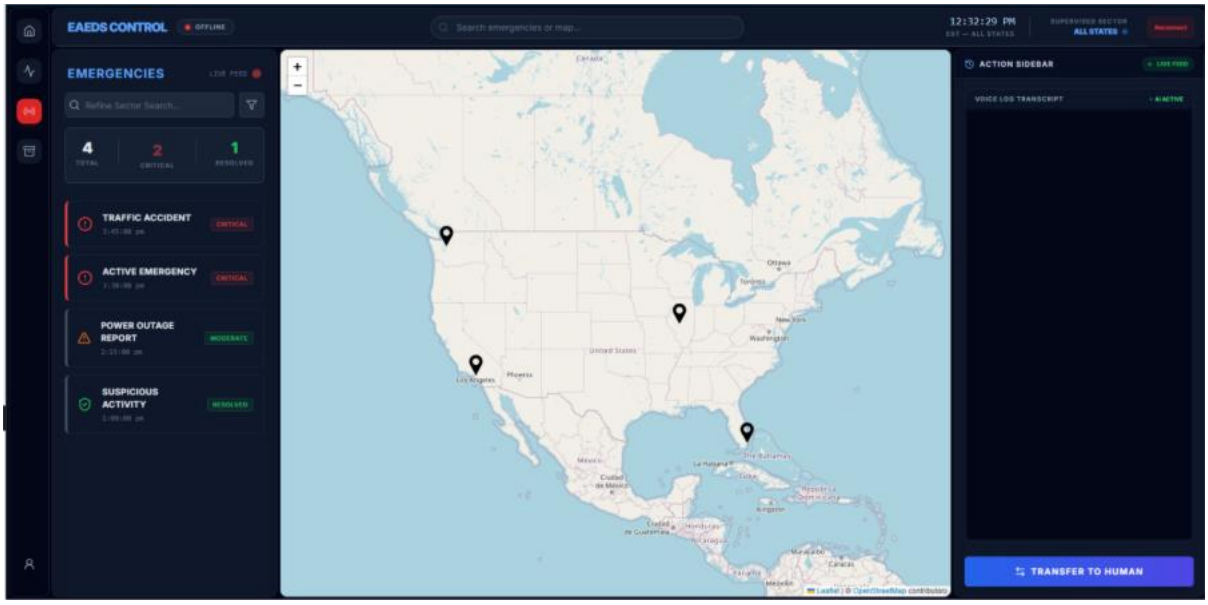
8.3 Phase III - Release Candidate

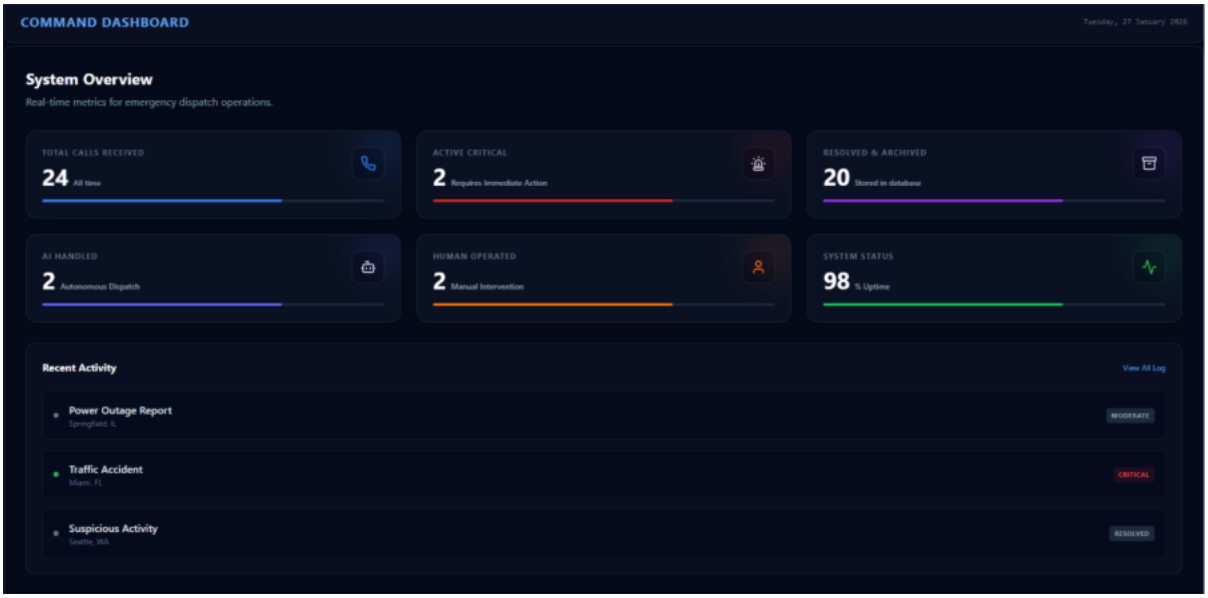
Integration of Local LLM (Mistral/Phi-3) for Dispatch Advice is fone
Full "Midnight Command" UI visualization is done.
Fail-safe manual modes is active.

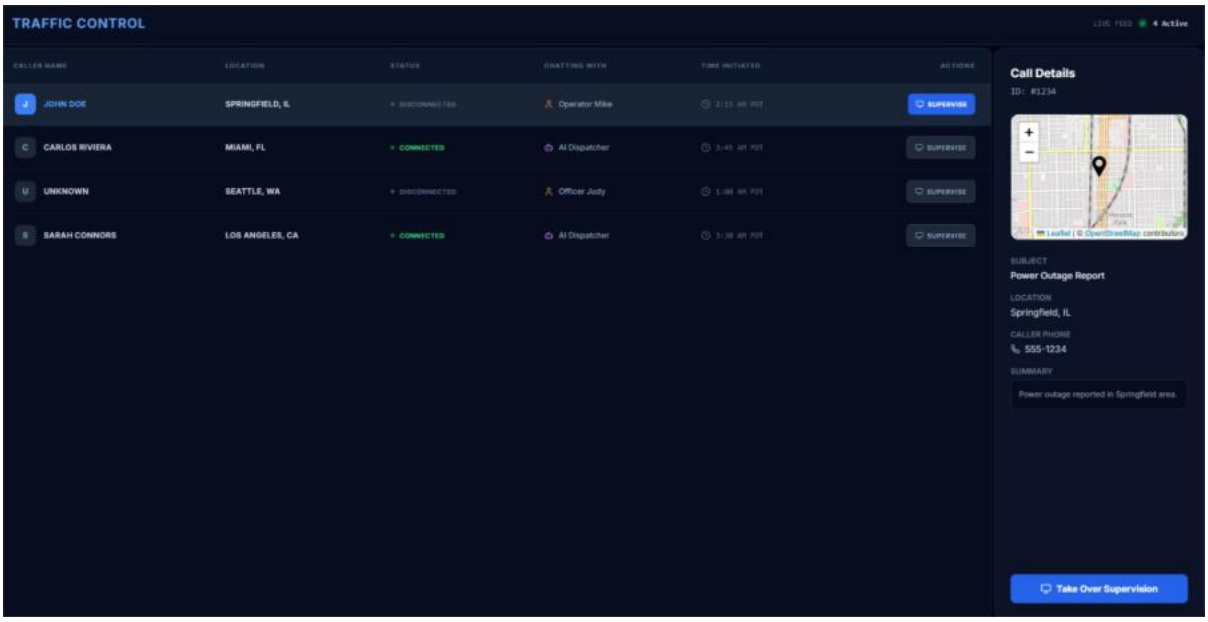
8.4 Phase IV - Final Product

Deployment package is fully dockerized
Comprehensive User Manual & Admin Guide.
Optimized Low-Latency performance tuning (<2s).

9 Proposed GUI (Prototype)






TRAFFIC CONTROL
4 Active
CALLER NAME
LOCATION
STATUS
CHATTING WITH
TIME INITIATED
ACTIONS
J JOHN DOE
SPRINGFIELD, IL
DISCONNECTED
Operator Mike
2:15 AM PDT
SUPERVISE
C CARLOS RIVIERA
MIAMI, FL
CONNECTED
AI Dispatcher
3:45 AM PDT
SUPERVISE
U UNKNOWN
SEATTLE, WA
DISCONNECTED
Officer Judy
1:00 AM PDT
SUPERVISE
S SARAH CONNORS
LOS ANGELES, CA
CONNECTED
AI Dispatcher
3:30 AM PDT
SUPERVISE
Call Details
ID: #1234
Leaflet | © OpenStreetMap contributors
SUBJECT
Power Outage Report
LOCATION
Springfield, IL
CALLER PHONE
555-1234
SUMMARY
Power outage reported in Springfield area.
Take Over Supervision

10 Meetings held with supervisor and/or client.

Meeting	Date	Key Discussion Points	Action Items	Deadline
1	1/7/2025	Project vision and objectives overview	Prepare initial project concept	8/7/2025
2	8/7/2025	Securing co-supervisor approval, alignment with expectations, project scope confirmation	Finalize initial project proposal draft	22/7/2025
3	22/7/2025	Tech stack discussion, documentation requirements planning	Begin working on SDS and SRS documents	6/8/2025
4	6/8/2025	SDS and SRS review, tech stack finalization (FastAPI backend, Next.js frontend), SQLite for call history storage	Complete detailed class diagram	20/8/2025
5	20/8/2025	AI models pipeline, UI requirements for main dashboard	UI prototype development	4/9/2025
6	4/9/2025	Real-time calling integration issues, frontend review, supervisor suggested using local calling instead of real landline	Integrate local calling system	18/9/2025
7	18/9/2025	SQLite finalized for efficiency, UI approved with suggestion to add total calls page showing critical calls	Add total calls tracking page	7/10/2025
8	7/10/2025	Dataset validation: 450 calls clean dataset in JSON format	Prepare data for models integration	28/10/2025
9	28/10/2025	Integration of Whisper, OpenSMILE, and BERT models, backend connection guidance	Integration of all AI models	14/11/2025
10	14/11/2025	Models integration review, discussion on starting fine-tuning process	Begin fine-tuning of models	25/11/2025
11	25/11/2025	Latency issues with Whisper, garbage values in input, LLM accuracy concerns, high GPU requirements. Supervisor suggested hybrid architecture and considering alternatives to Mistral LLM	Arrange high GPU or use cloud services	7/1/2026
12	29/1/2026	Final review of all documents before deadline, supervisor provided changes for final submission, presentation tips for February	Finalize all documents	30/1/2026

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